5.6 Composition of Functions

Name _____ Period ____ Date _____

Algebra II 5.6 · Student sheet

I can…

- evaluate a composition of functions.
- find the composition of two functions and its domain.
- use composition to solve real-life problems.

Vocabulary · fill in the missing word

Composition — f(g(x)): the _____ of g is the input of f

Domain of f(g(x)) — the _____ in the domain of g whose outputs g(x) are in the domain of f

Part 1 · Coupon vs. sale

A store has a 20%-off sale, s(x) = 0.8x. You also have a $10-off coupon, c(x) = x − 10.

Price	Sale first, then coupon	Coupon first, then sale
$60		
$80		
$100		

Which order is better for the shopper? Why?

Sale first, then coupon, as one function: c(s(x)) = _____ Coupon first: s(c(x)) = _____

Part 2 · Notes and examples (fill in each note, then work the example)

Composition: f(g(x)) means: find _____ first, then use that output as the _____ of f.

Composition: Work from the _____ out.

Example 1

Let f(x) = 5x − 1 and g(x) = x² + 2. Find f(g(−1)) and g(f(−1)).

Order matters: f(g(x)) and g(f(x)) are usually _____.

Example 2

Let f(x) = 3x − 4 and g(x) = x². Find f(g(x)) and g(f(x)).

Example 3

Let $f(x) = 2x + 7$. Find $f(f(x))$.

Domain of f(g(x)): The domain of $f(g(x))$: x must be in the domain of _____, and $g(x)$ must be in the domain of _____.

Example 4

Let $f(x) = \sqrt{x}$ and $g(x) = x - 5$. Find $f(g(x))$ and its domain.

Example 5

Use the table to find $f(g(0))$ and $g(f(-1))$.

x:	−2	−1	0	1	2
f(x):	3	1	−1	−3	−5
g(x):	1	−2	0	2	−1

Example 6

Use the coupon $c(x) = x - 10$ and the sale $s(x) = 0.8x$. Write $c(s(x))$ and find the price of an \$85 jacket with the sale first and then the coupon.

Part 3 · Partner practice

Partner 1

Let $f(x) = x^2 - 1$ and $g(x) = 2x + 3$. Find $f(g(x))$.

Partner 2

Let $f(x) = \sqrt{x + 1}$ and $g(x) = x^3$. Find $f(g(2))$.